

Emily Tran

Los Angeles, CA • (805) 248-9211 • emilyclaratran@yahoo.com • ectran.github.io/PortfolioWebsite

Education

California State Polytechnic University, Pomona

Bachelor of Science in Computer Science, Minor in Data Science

GPA: 3.74 • Magna Cum Laude

1st Place — Halloween Horror Game Jam (2025)

Aug 2022 – May 2026

Pomona, CA

Projects

Gambler's Addict | *Unity, C#, Gameplay Systems, Adaptive AI*

- Developed a 2D bullet-hell roguelike featuring adaptive betting, economy, and difficulty-scaling systems in Unity
- Engineered a score-weighted boss AI with dynamic attack selection and loop-based combat scaling
- Implemented persistent gameplay systems including progression tracking, adaptive multipliers, and cross-scene state management
- Developed debugging and visualization tools for gameplay balancing during playtesting

Funk vs Fright | *Unity, C#, Animation Systems, Gameplay Programming*

- Led a 4-person team in developing a 2.5D rhythm-combat game during a 10-day game jam
- Engineered timing-based combat mechanics and animation state systems in Unity
- Created original sprites, UI assets, and gameplay feedback elements
- Won 1st Place overall in the Halloween Horror Game Jam competition

CalendarMeet | *React, Node.js, PostgreSQL, AWS, Github Actions*

- Contributed to the development and deployment of a full-stack scheduling web application
- Built AWS deployment infrastructure and automated CI/CD workflows using GitHub Actions
- Deployed and maintained a React/Node.js/PostgreSQL application in a cloud-hosted environment

Experience

Virtual Reality Lab — Cal Poly Pomona

Unity Developer

Mar 2025 – Jun 2025

Pomona, CA

- Collaborated within a 4-person development team on a VR zombie survival FPS built in Unity using XR Interaction Toolkit
- Developed VR gameplay interaction systems, enemy AI behaviors, and spatial 3D audio systems
- Conducted playtesting, debugging, and iterative gameplay improvements for usability and immersion

Research Experience for Undergraduates (REU) — Texas A&M University

Applied Computational Robotics Researcher

May 2024 – Aug 2024

College Station, TX

- Developed Pygame simulations to evaluate autonomous rendezvous algorithms
- Implemented and analyzed A* and D* Lite pathfinding algorithms
- Conducted comparative testing and presented technical findings in a collaborative research environment

Technical Skills

Languages: C#, Java, Python, JavaScript, SQL

Game Development: Unity, Gameplay Systems, Game AI, Game UI, XR Interaction Toolkit, Pygame

Web & Cloud: React, Node.js, PostgreSQL, AWS, GitHub Actions

Tools: Git, GitHub, VS Code, Aseprite